

1. Crawler Task Data Sources: Yahoo,Finance,Reuters,CNBC

Crawling Strategy:

Search articles containing stock tickers or company names

Extract:Article headline,Full article text,Publication date,Source

Example

```
from newspaper import Article

url = "https://www.reuters.com/technology/apple-earnings-2025"
article = Article(url)
article.download()
article.parse()

headline = article.title
text = article.text
```

1. AI Task: Sentiment Analysis Gen AI Tool Chosen

OpenAI GPT-4 / GPT-3.5 (or Hugging Face FinBERT as an alternative)

Sentiment Classes:

Positive,Negative,Neutral

Example AI Output i Json

```
{
  "sentiment": "Negative",
  "keywords": ["earnings miss", "declining revenue", "weak guidance"]
}
```

1. Keyword Extraction Method

Keywords extracted directly by the Gen AI model

Focus on:

Financial performance

Earnings

Guidance

Legal / regulatory issues

Product launches

Daily Sentiment Aggregation (Per Stock)

For each stock:

Collect all articles published on the same day

Assign numeric values:

Positive = +1

Neutral = 0

Negative = -1

Compute average sentiment score

Insight Generation (AI-Generated)

Using Gen AI to convert sentiment trends into natural language insights.

Prompt Example Generate a one-sentence insight explaining the sentiment change based on earnings, product launches, or market events.

Sample Insights

AAPL: "Apple sentiment turned negative following weaker-than-expected earnings and reduced revenue guidance."

MSFT: "Microsoft sentiment remained positive due to strong cloud growth and optimistic forward guidance."

TSLA: "Tesla sentiment stayed neutral as delivery growth was offset by margin concerns."

GOOGL: "Alphabet sentiment improved after strong advertising revenue and AI product announcements."

META: "Meta sentiment turned positive following cost-cutting measures and improved profitability."

Output Format

```
{
  "date": "2026-02-10",
  "stocks": {
    "AAPL": {
      "sentiment": "Negative",
      "keywords": ["earnings miss", "weak guidance"],
      "insight": "Apple sentiment turned negative after earnings announcement."
    },
    "MSFT": {
      "sentiment": "Positive",
      "keywords": ["cloud growth", "AI expansion"],
      "insight": "Microsoft sentiment stayed positive due to strong cloud performance."
    }
  }
}
```